

Exercises

The first ten problems are Core Exercises. They contain the essential work for this chapter and are appropriate for all students. Problems 11–20 are Challenge Problems. They are optional and are intended especially for students who enjoy difficult problems or wish to prepare for MATHCOUNTS and the AMC 8. All problems in this section have been newly written for this book.

Part A. Core Exercises

1. Solve

$$4x - 9 = 2x + 7.$$

2. Solve

$$x - [6 - (x + 1)] = -5.$$

3. Solve

$$\frac{3x}{4} + \frac{5x}{8} = 22.$$

4. Solve

$$\frac{x}{3} + \frac{x}{5} = \frac{16}{15}.$$

5. Solve

$$5 - 2(x - 3) = 3x + 1.$$

6. Solve

$$\frac{2x + 7}{x + 4} = 3.$$

7. Solve

$$\frac{3}{x - 1} = \frac{5}{x + 1}.$$

8. Solve

$$(x - 2)^2 = (x - 5)^2 + 9.$$

9. Find m if $x = 3$ is a solution of

$$2(x - 1) = x + m.$$

10. Find m so that

$$4x - 2m = 4x + 6$$

has infinitely many solutions.

Part B. Challenge Problems

11. Solve

$$\frac{1 - \frac{2}{x}}{1 + \frac{2}{x}} = 3.$$

12. Solve

$$\frac{x-2}{x-5} = \frac{x+4}{x+1}.$$

13. Find
- m
- so that

$$(m-2)x + 5 = 3x - 1$$

has no solution.

14. Find
- m
- so that

$$(2m+1)x - 4 = (m+5)x + 7$$

has exactly one solution.

15. Without solving for
- x
- and
- y
- separately, suppose

$$3x - 2y = 7$$

and

$$5x + 2y = 13.$$

Find x .

16. Solve

$$\sqrt{9 + (x+2)^2} = \sqrt{49 + (8-x)^2}.$$

17. The equation

$$ax + 4 = 6x + b$$

has the solution $x = 2$ for every value of a . Express b in terms of a .

18. A number
- x
- has the property that increasing it by 20% gives the same result as adding 18 to it. Find
- x
- .

19. Consider the following two equations:

$$(k-1)x + 2k = 3x + 8,$$

$$(k-1)x + 2k = 3x + 9.$$

For each equation, find the value of k for which it does not have exactly one solution. Decide whether the exceptional case gives no solution or infinitely many solutions.

20. Consider

$$ax + b = cx + d.$$

Write a short summary, in your own words, explaining exactly when this equation has one solution, no solution, or infinitely many solutions. Then create one example of each type.

Beyond the Chapter

These questions are optional. Their purpose is not speed, but discovery.

1. Starting from

$$ax + b = cx + d,$$

derive the solution formula yourself without looking back at the chapter. Then explain why the formula fails when $a = c$.

2. Consider

$$x^2 = (a - x)^2 + c.$$

Derive a formula for x in terms of a and c . What special case occurs when $a = 0$?

3. Create a rational equation that looks nonlinear but simplifies to a linear equation. Solve it and explain why any excluded values must be checked.
4. Write three different equations that all have $x = 5$ as their unique solution. Try to make the equations look as different from one another as possible.

Answer Key

1. $x = 8$.
2. $x = 0$.
3. $x = 16$.
4. $x = 2$.
5. $x = 2$.
6. $x = -5$.
7. $x = 4$.
8. $x = 5$.
9. $m = 1$.
10. $m = -3$.
11. $x = -4$.
12. No solution.
13. $m = 5$.
14. Any $m \neq 4$ gives exactly one solution.
15. $x = \frac{5}{2}$.
16. $x = 5$.
17. $b = 2a - 8$.
18. $x = 90$.
19. In both equations the exceptional value is $k = 4$. The first equation has infinitely many solutions; the second has no solution.
20. Answers will vary. One solution if $a \neq c$; no solution if $a = c$ and $b \neq d$; infinitely many solutions if $a = c$ and $b = d$.